

FUTURE MISSIONS

▶ X-RAY POLARIMETER SATELLITE (XPOSAT)

India's first dedicated polarimetry mission to study various dynamics of bright astronomical X-ray sources in extreme conditions

▶ NASA-ISRO SAR (NISAR)

A Low Earth Orbit (LEO) observatory being jointly developed by Nasa and Isro to provide spatially and temporally consistent data by mapping the Earth in 12 days

▶ SPADEX

A space docking experiment involving two spacecraft to develop technologies related to orbital rendezvous, formation flying for applications in human spaceflight, in-space satellite servicing and other operations

▶ MANGALYAAN-2

A follow-up interplanetary mission to Mangalyaan-1 to study the early Martian crust

▶ GAGANYAAN MISSIONS

Comprising three missions, Isro will launch the country's first human spaceflight on Gaganyaan-3 after testing the first two unmanned launches

▶ SHUKRAYAAN

An orbiter will be sent to Venus as part of the interplanetary mission

▶ KULASEKHARAPATNAM SPACEPORT

Isro's second spaceport is being developed at Kulasekharapatnam in Tamil Nadu's Thoothukudi district for smaller launches

▶ SAMUDRAYAAN

Other than forays into outer space, Isro's Vikram Sarabhai Space Centre (VSSC) is also developing a titanium alloy submersible for a 6,000-metre-deep undersea crewed mission in association with the National Institute of Ocean Technology (NIOT)

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rocket, and we would like to transfer this rocket to the industry at the earliest. The industry will be given the whole technology to build and operate it to support small satellite constellations at manageable costs," Isro Chairman Somanath tells *Business Today*.

Following a process initiated by the Indian National Space Promotion and Authorisation Centre (IN-SPACE)—the single-window nodal agency for authorising, promoting and regulating private players in the space sector—a total of 23 private players have expressed interest in building the SSLV launch vehicle after the technology transfer

from the national space agency.

Somanath has also sought the private sector's support in expediting its reusable rocket programme to compete with Elon Musk's SpaceX. He has already committed Isro's support in working with the private sector in designing the complex technology that will go into building the reusable rockets. "The rocket has to be owned and operated by business houses to not only handle more payloads commercially but also be able to increase its numbers whenever required. This vehicle needs to be realised in the shortest possible time because time is running out," cau-

tions Somanath. His sense of urgency can be gauged from the successes that SpaceX has had with the Falcon 9 rocket, in terms of bringing down the cost of launching payloads in LEO to as low as \$2,700 per kg. And a successful deployment of its under-development Starship will see this crashing further to as low as \$10 per kg.

But looking at the list of its overly ambitious space missions planned as well as the can-do attitude, Isro may spring a surprise with a far cheaper, yet more reliable, alternative. **BT**

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